PRACTICE AND ASSESSMENT

Lesson 7 Exercises and Explained Solutions

SVD, PCA, low-rank approximation and responsible interpretation

SRAI - Statistics, Reasoning and Artificial Intelligence

Book 1: Mathematical Foundations | Lesson 7

Instructions

Complete the reasoning, calculations and interpretation. Numerical answers should include tolerances and reproducible evidence. Governance answers must identify what evidence is insufficient.

1. SVD reconstruction

For A = [[3,1],[1,3]], compute or obtain U, Σ and V^T. Reconstruct A and verify both orthogonality identities.

2. Rank-one approximation

Construct the rank-one approximation of the same matrix and calculate its Frobenius reconstruction error.

3. Choosing k

Explain why choosing k solely because cumulative explained variance exceeds 90% can be unsafe.

4. Centering

What happens if PCA is applied without centering?

5. Standardization

When should variables be standardized before PCA?

6. Signs

Two software packages return opposite signs for PC1. Is one wrong?

7. Loadings

A component loads positively on electricity, health and transport. May it be called “development”?

8. Leakage

Describe a leakage-safe train/test PCA workflow.

9. Equity audit

Name three checks before deploying compressed indicators in a public-sector model.

10. Decision statement

Write a preservation-and-loss statement for a two-component national-indicator model.

Explained solutions and marking guidance

1. SVD reconstruction

A numerical SVD has singular values 4 and 2. Multiplying $U\Sigma V^T$ returns A to floating-point tolerance. Verify $U^TU = I$ and $V^TV = I$. Sign-flipped singular-vector pairs are equivalent.

Suggested marking: 4 - correct and fully reasoned; 3 - correct with limited justification; 2 - partial structure; 1 - relevant attempt; 0 - absent or unrelated.

2. Rank-one approximation

Retain only σ_1 and its singular-vector pair. The discarded singular value is 2, so the optimal rank-one Frobenius error is 2.

Suggested marking: 4 - correct and fully reasoned; 3 - correct with limited justification; 2 - partial structure; 1 - relevant attempt; 0 - absent or unrelated.

3. Choosing k

Variance measures reconstruction, not harm or decision importance. Low-variance directions can encode minority groups, anomalies or thresholds. Choose k using reconstruction, downstream performance, subgroup checks and decision consequences.

Suggested marking: 4 - correct and fully reasoned; 3 - correct with limited justification; 2 - partial structure; 1 - relevant attempt; 0 - absent or unrelated.

4. Centering

The origin rather than the sample mean anchors the directions. Components may reflect the mean offset instead of covariance structure, so standard PCA interpretation is invalid.

Suggested marking: 4 - correct and fully reasoned; 3 - correct with limited justification; 2 - partial structure; 1 - relevant attempt; 0 - absent or unrelated.

5. Standardization

Standardize when scale differences are measurement artifacts or when comparable variance weighting is intended. Do not standardize automatically when absolute scale carries legitimate substantive meaning.

Suggested marking: 4 - correct and fully reasoned; 3 - correct with limited justification; 2 - partial structure; 1 - relevant attempt; 0 - absent or unrelated.

6. Signs

Not necessarily. A direction and its negative span the same axis. Compare subspaces, explained variance and reconstructed values rather than raw signs.

Suggested marking: 4 - correct and fully reasoned; 3 - correct with limited justification; 2 - partial structure; 1 - relevant attempt; 0 - absent or unrelated.

7. Loadings

Only with independent domain evidence. Loadings show mathematical association. A substantive label should be justified, bounded and tested against excluded variables and alternative interpretations.

Suggested marking: 4 - correct and fully reasoned; 3 - correct with limited justification; 2 - partial structure; 1 - relevant attempt; 0 - absent or unrelated.

8. Leakage

Split first. Fit centering, scaling and PCA only on training data. Apply the fitted transformations to validation/test data. Never refit using test observations.

Suggested marking: 4 - correct and fully reasoned; 3 - correct with limited justification; 2 - partial structure; 1 - relevant attempt; 0 - absent or unrelated.

9. Equity audit

Compare reconstruction error by subgroup; inspect rare or high-impact observations; test downstream performance and decisions with and without compression. Also document ownership and escalation thresholds.

Suggested marking: 4 - correct and fully reasoned; 3 - correct with limited justification; 2 - partial structure; 1 - relevant attempt; 0 - absent or unrelated.

10. Decision statement

Example: The two components retain the dominant shared variation among five standardized service indicators and reproduce most regional profiles. They remove residual patterns, including possible sector-specific and subgroup differences; original indicators remain available, residuals are reviewed by region, and compression is not used alone for resource allocation.

Suggested marking: 4 - correct and fully reasoned; 3 - correct with limited justification; 2 - partial structure; 1 - relevant attempt; 0 - absent or unrelated.